Can Agents Trust a Replication Verifier? Epistemic Contracts for Scientific Agent Tools

Luxin Zhang
Independent Researcher
luxin.zhang92@outlook.com

***Abstract*—Agentic AI systems can delegate the question "is this scientific claim trustworthy?" to a verifier tool, yet such tools are evaluated on accuracy rather than on the guarantees a calling agent relies on. We propose a testable methodology for *epistemic contracts* for agent tools: guarantees about what an output may reflect (temporal validity, label isolation), what decisions it licenses, and how it is audited. We apply it to SCIRELIGRAPH, an MCP replication verifier whose contract tests reach a verified mutation score of 61% (83% after survivor-motivated tests). On the 274 SCORE claims with replication outcomes, an integrity-compliant verifier discriminates only modestly (AUROC 0.627), and bibliographic evidence graphs add no established improvement. The best of four LLM judges rank competitively, but their temporal validity cannot be certified, and a decoy-calibrated probe shows two judges naming outcomes for never-replicated papers. In an agent-in-the-loop experiment with no-tool and placebo controls, a bare probability induced near-chance automatic decisions that most agents avoided without the tool; the contract removed most of them, beyond a generic warning, for three of four agents, and the fourth misread it. In a pre-specified follow-up, wordings that state the licensed action eliminated misuse for all four agents (three also under a second prompt), and with a synthetic decision-ready verifier the contract let agents automate exactly the licensed claims (4.7% wrong vs. 12–15% with a bare number). Contracts thus convey when automation is warranted, and should be tested against the agents that consume them.**

***Index Terms*—agentic AI, epistemic contracts, AI for science, verification, LLM-as-a-judge, contract testing, metamorphic testing, data leakage, MCP, replicability**

I. INTRODUCTION

Systems that automate parts of research [1] must repeatedly decide which prior findings are solid enough to build on, and can delegate that decision to a *verifier* tool such as an LLM-as-a-judge [2]. The decision matters: many published social and behavioural science findings do not replicate [3], [4], [5], and the DARPA SCORE program released outcomes for 274 replicated claims from 164 papers, of which 151 (55.1%) replicated [6], [7]. Delegation also warrants caution: LLMs that predict experimental outcomes cannot tell their reliable predictions from unreliable ones [8], and AI-led teams reproduced far fewer published results than human-only or AI-assisted teams [9].

An agent calling a verifier at time t about claim c assumes that (i) the answer uses only information available at t, (ii) the probability means what it says, and (iii) the answer can be audited. We call such guarantees an *epistemic contract*: unlike security and correctness checks, it states what the tool's *output may reflect* and what decisions it *licenses*. Its ingredients—design by contract [10], output documentation [11], calibration [12], metamorphic [13] and mutation testing [14], tool-use evaluation [15]—are established; our contribution is a *testable methodology* that combines them for agent tools, and evidence of a failure mode visible only when they are combined. The guarantees fail silently: a verifier fitted on bibliometric data scraped today can absorb post-replication signals, an LLM judge may remember the outcome [16], and random splits put one paper's claims in both training and test data [17], [18].

We ask: *if a verifier is built and tested against an explicit epistemic contract, is it good enough for an agent to rely on, and does the agent use it correctly?* Our hypotheses that time-bounded evidence graphs (H1), corrected for non-independent support (H2), would improve forecasts, and that lifting the cutoff would inflate accuracy (H3), were not supported; we report the negative results in full. The agent experiment motivates testing the *consumer* of the contract: one agent read the tool's honest warning that its score was weak as evidence *against the claim* and rejected 96% of claims (§VII).

Contributions.

1) *A testable epistemic-contract methodology for verifier tools* (§III): five requirements, a feature-availability typing that makes temporal validity checkable, and a reusable machine-readable specification (a JSON Schema for answers and refusals, with tool-agnostic test relations). In our own pipeline, the typing caught a leak that would otherwise have produced our "best" harness result.
2) *Contract tests with verified adequacy* (§IV–§V): an MCP tool that enforces the contract at call time, metamorphic and isolation tests, and a two-level mutation analysis (targeted mutants, and a standard tool whose verdicts we re-verified independently).
3) *A decoy-calibrated memorization probe for LLM judges* (§VI-F). Asking judges about journal-matched papers that were never replicated or reproduced separates recall from fabrication: two of four models named outcomes for such papers.
4) *Evidence of a contract–consumer gap* (§VII). With no-tool and placebo controls, a bare verifier probability induced errors that most agents did not make without the tool; the contract removed most of them, beyond a generic warning, for three agents, and misled a fourth. A pre-specified follow-up shows that stating the licensed

TABLE I

POSITIONING AGAINST THE CLOSEST WORK, BY EACH WORK'S STATED SCOPE. ✓: A STATED FOCUS; ○: ADDRESSED IN A DIFFERENT SENSE; –: NOT A STATED FOCUS (NOT A CLAIM OF ABSENCE). OUR AGENT EVALUATION IS SINGLE-TURN, HENCE ○ IN (E). COLUMNS: (A) REALISED OUTCOMES AS LABELS; (B) TEMPORAL LEAKAGE CONTROLLED; (C) CALIBRATION OR SELECTIVE ACCURACY EVALUATED; (D) THE TOOL'S OUTPUT CONTRACT TESTED; (E) AGENT USE OF THE OUTPUT EVALUATED.

Work	(a)	(b)	(c)	(d)	(e)
Replication predictors [22], [23], [24], [26]	✓	–	–	–	–
SciPredict [8]	✓	✓	✓	–	–
GLEAN [32], CLAIM-CAL [33]	✓	–	✓	–	○
ReplicatorBench [28]	✓	–	–	–	✓
MCP agent benchmarks [15], [34], [36]	–	–	–	○	✓
MCP runtime defence [35]	–	–	–	○	–
This work	✓	✓	✓	✓	○

action fixes this for all four agents (three also under a second prompt), and that the same contract licenses automation when a tool is decision-ready.

Empirically, on the full SCORE replication set, the best leakage-safe verifiers are weak (AUROC ≈0.63), bibliographic evidence graphs yield no statistically established improvement, and the best LLM judges are competitive but uncertifiable. Replication forecasts should therefore inform agents, not gate them.

II. Background and Related Work

Replication prediction. Prediction markets classified 103 replication outcomes with 73% accuracy [19], [20]; for SCORE, human forecasters reached a median AUC of 0.76 on 37 high-powered replications [21]. Machine-learning forecasters using statistical and textual features report above-chance accuracy [22], [23], [24]; SCORE's machine assessments mostly targeted *human* judgements [25], [26]; hybrid markets add algorithmic traders [27]; and ReplicatorBench evaluates agents that *carry out* replications [28]. We differ in using only realised outcomes as labels, enforcing temporal validity, and evaluating the verifier as a tool an agent acts on. Nonreplicable findings are cited more [29]; we test whether citation *structure* helps where counts do not. Unlike fact-checking [30], replication forecasting predicts a future study and is exposed to hindsight leakage.

LLM judgements and agent tools. LLM forecasters are vulnerable to temporal leakage [31]. GLEAN calibrates trajectory evidence for high-stakes agent verification [32], and claim-level calibration applies the same post-hoc calibration to every baseline [33], as we do. MCP work evaluates multi-step tool use [15], metamorphic exploration of tool spaces [34], run-time defences [35] and agents' actions in experiment reproduction [36]. These ask whether agents *use* tools correctly or safely; we ask whether a tool's *output* keeps a contract the agent can rely on, and whether agents consume it correctly (Table I). Hidden data dependencies are a known source of ML technical debt [37], [38], datasheets make provenance explicit [39], and MCP [40] makes it easy to expose a predictor as a tool but not to specify what its output may be trusted for.

III. A Verifier Contract for Agentic Pipelines

A. Problem definition

Let c be an original empirical claim, t a forecast cutoff, and $\mathcal{E}(c,t)$ the admissible evidence dated $\leq t$. Let $Y_c \in \{0,1\}$ be the outcome of a later independent replication under a pre-declared criterion. The verifier returns $\hat{p}_c = \hat{P}(Y_c = 1 \mid c, \mathcal{E}(c,t))$ with an evidence record. We evaluate discrimination (AUROC), probabilistic accuracy with strictly proper scores (Brier, log loss) [41], calibration [12], and selective accuracy [42]. The last of these matters because an agent may act only on confident outputs.

B. Requirements of the epistemic contract

R1 Temporal validity. No feature or evidence of the queried claim may encode information dated after t, in particular replication results, replication-paper metadata, or post-cutoff citations. (The model is necessarily trained on *other* claims' later outcomes, and the reported base rate is itself post-t; we treat this residual exposure as a limitation.) Calls whose t falls after the outcome may be public are refused. **R2 Label validity.** The label is the declared criterion, reconciled to published counts; alternative criteria are reported as sensitivity analyses. **R3 Honest generalisation.** Claims from the same paper are never split across training and test data, and a claim is never scored by a model that saw any label from its own paper. **R4 Decision-relevant output.** The output must beat the base rate on proper scores and must expose the base rate. If it is used for gating, its confident outputs must be more accurate than its unconfident ones. **R5 Auditability.** Every prediction must be traceable to a source archive (with checksum), an evidence access date, the feature tiers used, and a model configuration. R1 and R3 constrain what the output may *reflect*, R4 what it *licenses*, and R2 and R5 make both checkable. R1, R2, R3 and R5 are enforceable integrity requirements. R4 is different: it is an empirical acceptance criterion that a verifier may fail. A contract-compliant response must then *expose* that failure, through its base rate, skill and gating verdict, rather than imply that the predictor is decision-ready. We therefore distinguish three levels: a verifier is *integrity-compliant* if it meets R1–R3 and R5, *decision-qualified* if it also meets R4, and *consumer-validated* if its output has been tested against the agents that consume it (R6, §VIII). In our results, R1–R3 and R5 are met by construction and tested (§V). R4's first clause is met only nominally (B1 Brier skill 2.6%, paired CI includes zero), and its gating clause is not met by any harness verifier (§VI-E).

C. Feature-availability typing

R1 is enforced by *typing* every feature with the moment it becomes knowable. **Tier A** (original publication): sample size; p-value and whether it is reported exactly; marginal significance ($0.01 < p \leq 0.05$); converted effect size $|r|$; power to detect half the original effect; interaction vs. main effect; test-statistic and analysis family; publication year; and discipline. **Tier T** (original publication): title, abstract and keywords. **Tier G(t)** (cutoff-dependent): the evidence graph

TABLE II
REQUIREMENTS, ENFORCEMENT MECHANISMS, AND WHAT ENFORCEMENT CAUGHT OR SHOWED IN THIS STUDY.

Req.	Enforcement mechanism	Finding in this study
R1 temporal	Feature tiers; dated-evidence filter; assertion that the cutoff precedes the earliest replication start; call-time refusal; metamorphic test	Excluded TOP Factor scores (undated); exposed the achieved-size leak (Tier D*)
R2 label	Build-time assertion of 274/164/151; alternative criteria kept as separate labels	Exact reconciliation; three sensitivity labels
R3 generalisation	Paper-grouped CV; nested grouped tuning; leave-paper-out in the tool; label-flip test	Unregularised starter baseline scored worse than the base rate
R4 decision use	Proper scores vs. base rate; reliability diagram (artifact); selective accuracy; `selective_ok` flag	B1 Brier skill 2.6% (CI includes 0); no harness verifier shows a supported gating gain
R5 audit	Manifest (sha256, access date, tiers, missingness); per-call audit log; scripted tables and numbers	All reported numbers regenerate from outputs

of §IV. **Tier D** (replication launch): new vs. secondary data, the preregistered power target, and the *planned* sample size implied by that target. **Tier D*** (after data collection): the *achieved* replication sample size and the power computed from it. A forecast "at publication" may use A, T and G(t); a forecast "at launch" may add D. D* is never admissible and appears only as a leakage illustration: the achieved size exceeds the planned size by a median factor of 1.33 and can depend on how data collection went. Typing also forced decisions about fields whose timing the archive does not record. SCORE's journal-level TOP Factor policy scores look like publication metadata, but their rating date is not stored, so R1 cannot be verified for them and we exclude them.

IV. THE SCIRELIGRAPH HARNESS

Table II maps each requirement to its enforcement point: the label gate (R2), the evidence cutoff (R1) and the tool boundary (R1, R3–R5). The implementation is 2822 lines of Python and runs end to end from public sources.

Data and E0 gate. The harness downloads the publication-frozen SCORE analyst package [7] and records its SHA-256. It selects version-of-record, non-COVID replications and *asserts* that they reproduce the published totals of 274 claims, 164 papers and 151 successes [6]. Of the plausible selections we tried (e.g., 427 reports, 309 claims including COVID-era studies), only this one reproduces the publication. The set spans 54 journals, publication years 2009–2018 and six fields; 128 replications used new data and 146 secondary data.

Evidence and cutoffs. For each original paper, resolved by DOI (164/164), the harness caches the OpenAlex [43] record and every citing work: 23,666 in total, with authorships, venues, topics and references. Time filtering is applied only at feature construction. The *strict* cutoff, 2018-12-31, is asserted to precede the earliest replication start in the archive (2019-08-06); 8,268 citing works fall before it. The *launch* cutoff is the day before each replication project started; it is known for 259 claims, and the remainder use the strict cutoff. The *none* cutoff admits everything available at access time (October 2026) and is a diagnostic only.

Evidence graph G(t). For claim c, the graph contains the original paper and all citing works dated $\leq t$, with *cites*, *shares-author* (dependency) and *shares-reference* (coupling) edges. We derive (i) flat evidence (citation counts and rates, reference count); (ii) graph structure (citing-venue entropy, field diversity, cross-field citer share, citer–citer density, coupling); and (iii) dependency correction (self-citation share, independent support groups—connected components of non-self citers under shared authorship—and distinct citing authors). The graph is bibliographic (no semantic supports/contradicts edges), and its features are paper-level.

Models and protocol. B0 predicts the training-fold prevalence. Tabular models use median imputation with missingness indicators, standardisation and one-hot encoding (rare levels pooled), followed by L2 logistic regression. The regularisation strength $C \in \{0.003, \ldots, 1\}$ is chosen by inner grouped 3-fold CV on log loss. A gradient-boosted tree baseline (depth 2, 150 iterations) has fixed hyperparameters. The text model is TF-IDF (1–2-grams) with tuned logistic regression. We use stratified, paper-grouped 5-fold CV repeated over 20 seeds, and report the mean and SD over repetitions. For inference we average each claim's out-of-fold predictions over seeds and draw 2,000 paper-clustered bootstrap resamples. Paired differences are computed on identical resamples, and their p-values are Holm-adjusted over the family of 11 planned comparisons; reported CIs are unadjusted, so a CI can exclude zero while p_{Holm} is large. The **permutation null** permutes all graph and dependency features across papers within publication-year strata, 30 times, re-running the full CV each time. This preserves marginal distributions and the year–citation relation while breaking the link to outcomes.

V. THE VERIFIER AS AN AGENT TOOL

Offline evaluation establishes how good a verifier is; it does not ensure that the deployed tool behaves as evaluated. We therefore implemented the verifier as a library with an MCP [40] server exposing a single tool, `verify_claim(claim_id, as_of, stage)`, and enforced the contract inside the call path. (1) *R1:* for every call, the graph features of *all* claims are recomputed from the raw cache with evidence filtered to `as_of`, and calls on or after the claim's replication launch are refused. (2) *R3:* the model is refitted without any claim from the queried paper (leave-paper-out), with the same nested tuning as the harness. (3) *Stage gating:* `publication` uses tiers A+G, and `launch` uses A+D. (4) *R4:* the response carries the base rate, the cross-validated AUROC and Brier skill of that configuration, and a `selective_ok` flag that is true only if the lower 95% bound of top-decile accuracy exceeds overall accuracy in the selective-accuracy test (§VI-E); it is false for every harness configuration we evaluated. (5) *R5:* each call appends a record with inputs, outputs, archive hash and model-configuration hash to an audit log. Listing 1 shows the actual responses an MCP client received in our smoke test: one valid call and one refused call.

Listing 1. Actual `verify_claim` responses over MCP stdio (truncated hashes).

```
// verify_claim("Ej3y_single-trace", as_of="2018-12-31")
{"claim_id": "Ej3y_single-trace", "stage": "publication", "
    as_of": "2018-12-31",
 "p_replicate": 0.4673, "base_rate": 0.5531, "cv_auroc":
     0.62,
 "cv_brier_skill": 0.0277, "selective_ok": false, "
     n_evidence": 8,
 "tiers_used": ["A", "G"], "n_train_claims": 273, "drivers":
      [{"feature": "g_cross_field_frac", "
     logit_contribution": 0.119}],
 "provenance": {"archive_sha256": "80a221f0deb7...", "
     model_config": "81f3fdc8b3c61867"}, "advice": "weak
     signal: do not gate actions on this score"}
// same claim, as_of="2023-01-01"
{"error": "contract_violation", "detail": "as_of 2023-01-01
    is on/after replication launch 2021-05-24: evidence may
     reflect the outcome"}
```

Contract tests and their adequacy. The "right" probability is unknown, so we test *relations* that must hold under the contract (Table III): a *metamorphic* test injects citing works dated after `as_of` into every paper's evidence and requires a bit-identical output (a positive control checks that earlier works *are* counted); an *isolation* test flips the labels of every claim from the queried paper and requires an unchanged prediction; a *gating* test randomises Tier-D values and requires an unchanged publication-stage output; further tests check refusals, malformed calls, the base rate and audit records. A JSON Schema, adoptable by any verifier, fixes the response shape, and schema-conformance tests check answers and refusals against it. We built 5 mutants, each removing one guarantee (evidence filtering, leave-paper-out, stage gating, the launch refusal, audit logging); the unmodified tool passes all 9 original tests and every mutant is killed (5/5).

Independent mutation analysis. Because we wrote these mutants ourselves, we also ran `mutmut` 2.5.1 over the verifier module (251 mutants; Table IV). Its own verdicts missed some kills, so we re-ran every reported survivor in isolation with a freshly restored source file; 19 were in fact killed. The remaining survivors exposed real gaps, closed in two rounds: ten tests for the exact output schema, the true branch of the gating rule, evidence and training counts, and audit hashing; then four for the temporal boundary R1 depends on (a work dated exactly t is admitted, one dated $t{+}1$ excluded) and two oracles (the launch-date fallback; loaded evidence matches the raw cache). These fourteen tests were written after seeing the survivors, so the final score is optimistic; we report it alongside the pre-analysis 61%. None of the 42 remaining survivors threatens a contract guarantee: 13 change hyperparameters the contract leaves open, 9 touch the command-line entry point, 8 alter the untested "drivers" output, 5 change wording or rounding, and 7 are equivalent or near-equivalent.

VI. Results on the SCORE Testbed

Replication forecasting serves here as the testbed for the contract: it supplies realised outcomes, a natural temporal boundary, and a decision an agent would plausibly delegate. We first ask how good an integrity-compliant verifier is (A–E), then how LLM judges behave under the same contract (F), and finally check robustness (G).

A. How good is a careful verifier?

Table V summarises the main configurations. The publication-time metadata model B1 reaches AUROC $0.627{\pm}0.017$ (CI [0.55, 0.71]; B0's AUROC below 0.5 is an artefact of grouped CV, which shifts the training prevalence opposite to each fold's own). Refitting the whole protocol inside 200 paper-bootstrap resamples, which adds fitting and tuning variability, widens B1's interval to [0.46, 0.72] (93% of replicates above 0.5). That interval is conservative, since each resample holds only about 63% distinct papers, but the evidence of signal is fragile. The probabilistic gain is small: Brier 0.2409 vs. 0.2474, a skill score of 2.6%, with a paired CI that includes zero. A gradient-boosted variant ranks similarly (0.620) with a worse Brier score (0.2483); the TF-IDF text model is at chance (0.481). Replicated claims had stronger original evidence (mean $-\log_{10} p$ 8.9 vs. 4.7), higher power (0.55 vs. 0.44) and fewer marginal p-values (16% vs. 27%), in line with the replication literature. An unregularised starter baseline scored *worse* than the base rate (Brier 0.267 vs. 0.247), so base-rate comparison belongs in a verifier's acceptance tests.

B. H1/H2: Do evidence graphs and dependency correction help?

Not detectably. Flat citation evidence added to metadata changes AUROC by -0.013 (unadjusted CI $[-0.027, -0.001]$; $p_{\mathrm{Holm}} = 0.34$ after adjustment for 11 comparisons). Graph structure (M1 vs. B4: $+0.013$, CI $[-0.028, +0.046]$) and dependency correction (M2 vs. M1: $+0.000$, CI $[-0.017, +0.017]$) are null, and M2 is indistinguishable from metadata alone ($+0.000$, CI $[-0.041, +0.033]$). In the permutation null (30 within-year permutations, 5 seeds), the real graph block reaches 0.632 against a null mean of 0.605 (range 0.580–0.640; $p = 0.10$): perhaps weak signal, but not enough to offset the variance it adds (B1: 0.639 on the same seeds). Graph features alone reach 0.591, and raw pre-cutoff citation counts are uninformative (0.52), in line with [29].

C. H3: Does removing the cutoff inflate performance?

Not detectably for these features. Admitting all evidence available in 2026 raises the mean $\log(1+\text{citations})$ from 3.30 to 4.50 but does not improve discrimination ($\mathrm{M2}_\infty$ vs. M2: -0.010, CI $[-0.059, +0.035]$; G_∞ vs. G: -0.030, CI $[-0.107, +0.050]$). With so little signal this test has low power; it does not license dropping the cutoff, and it says nothing about LLM verifiers [16].

D. Launch-time information and a leak we caught

Moving the cutoff to each replication's start date leaves performance unchanged ($\mathrm{M2}_\ell$: 0.620). Our first "launch-time" model used the archive's replication sample size and power and appeared best; typing revealed these are *achieved* values, known only after data collection. With the planned design only

TABLE III
CONTRACT TESTS FOR THE VERIFIER TOOL AND MUTATION ANALYSIS. EACH MUTANT REMOVES ONE GUARANTEE FROM THE IMPLEMENTATION; × MARKS THE TESTS THAT FAIL (KILL THE MUTANT). THE UNMODIFIED VERIFIER PASSES ALL 9 TESTS.

Req.	Contract test	leaky	no-LPO	no-gate	no-refuse	no-audit
R1	Metamorphic: citing works dated after `as_of` added to every paper leave output unchanged	×				
R1	Positive control: works dated before `as_of` are counted					
R1	Call on/after the replication launch is refused				×	
–	Malformed date or unknown stage is refused (2 cases)					
R3	Flipping all labels of the queried claim's paper leaves its prediction unchanged		×			
R1	Publication stage: Tier-D values have no influence			×		
R4	Base rate equals training prevalence; weak-signal advice when selective test fails		×			
R5	Each call appends an audit record with provenance hashes					×
Mutant killed?		yes	yes	yes	yes	yes

TABLE IV
MUTATION SCORE OF THE VERIFIER MODULE (`mutmut`, 251 MUTANTS), BY TEST SUITE. EACH VERIFIED ROW RE-RUNS ALL SURVIVING MUTANTS IN ISOLATION. THE THREE LATER SCHEMA-CONFORMANCE TESTS WERE NOT PART OF THESE RUNS.

Stage	Tests	Killed	Score
Tool's own report	9	134	53%
Verified: original contract tests	9	153	61%
+ round 1 (schema, gating rule, counts, audit)	19	204	81%
+ round 2 (cutoff boundary, oracles)	23	209	83%
excluding 7 equivalent mutants	23	209	86%

(Tier D), the gain disappears (D vs. B1: -0.003, CI [-0.037, $+0.034$]); adding the achieved values back (D*) recovers part of it ($+0.024$, CI [-0.005, $+0.053$]), not significantly, but in the direction post-hoc information would produce. A field's *name* ("replication sample size") does not reveal its availability time; the typing step did.

E. Can an agent gate on confidence?

An agent may act only when a verifier is confident. Ranking claims by $|p - 0.5|$, B1's most confident 10% are correct 0.44 of the time (CI [0.26, 0.68]), versus 0.63 overall (D: 0.48 vs. 0.64). M2's top decile (0.67 vs. 0.59) has an interval [0.38, 0.82] far too wide to support gating, and mean predictions per quintile span a narrow range (reliability diagram in the artifact). R4's gating clause is therefore not met by any harness configuration; §VII shows what this means under a review budget.

F. LLM-as-judge verifiers

Agents can also delegate this decision to an LLM, so we evaluated four LLMs from two vendors under the same scoring protocol. Each vendor contributed one model whose training data end before the SCORE results paper (April 2026 [6]) and one after it (vendor-stated cutoffs): `gpt-5.4-mini-2026-03-17` (August 2025) and `gpt-6.1-sol` (April 2026), `claude-haiku-4-5-20251001` (training data to July 2025) and `claude-opus-5-5` (June 2026). Each model received a system prompt dated at the strict cutoff and asking for the probability that a high-powered replication would succeed, as JSON. We used one sample per claim and each model's lowest available reasoning setting. There were three conditions: (i) *identified*: title, citation, abstract and the Tier-A statistics; (ii) *de-identified*: field and statistics only, which removes the cues needed to recognise the paper; (iii) a *memorization probe* asking directly whether the paper was replicated in SCORE and with what outcome, also asked about 100 *decoys*: journal-matched papers that SCORE sampled but never replicated or reproduced. All responses parsed; no request was refused. The judging calls cost $5.68 in total.

Discrimination and probabilistic accuracy (Table VI). The two post-cutoff models achieved the highest observed AUROCs of all verifiers in this study (GPT-6.1 sol 0.663, CI [0.58, 0.74]; Claude Opus 5.5 0.655, CI [0.57, 0.74]), without a statistically established advantage over B1 ($+0.036$ and $+0.027$). The pre-cutoff models were below B1 (0.600 and 0.566). Raw probabilities differ sharply in quality. Opus 5.5's raw Brier score is as low as B1's (0.2392 vs. 0.2387; Brier mixes ranking and calibration, so this does not establish equal calibration), whereas the other three are worse than the base rate (0.2594, 0.2753, 0.2541; B0 0.2474). Following [33], we applied the same in-fold Platt scaling to every judge. This brings GPT-6.1 sol to 0.2313 and Opus 5.5 to 0.2323 (Brier skill +6.5% and +6.1%), the lowest of any verifier, while leaving AUROC almost unchanged. Neither recalibrated judge is distinguishable from B1 (ΔBrier CI [-0.0229, $+0.0074$] and [-0.0258, $+0.0132$]). The pre-cutoff judges improve only marginally (0.2466 and 0.2438 vs. B0 0.2474). Because a monotone recalibration improves their Brier scores without changing their ranking, the post-cutoff judges' probabilistic failure is mainly one of calibration. Recalibration, however, requires labelled outcomes, so a recalibrated judge is no longer zero-shot.

Memorization and fabrication. Removing identifying information lowers the point estimate of AUROC for both post-cutoff models (GPT-6.1 sol $+0.051$, CI [$+0.008$, $+0.101$]; Opus 5.5 $+0.035$, CI [-0.024, $+0.092$]; these LLM comparisons were not in the planned Holm family). The pattern is suggestive, not established: the pre-cutoff GPT-5.4 mini shows a similar drop ($+0.030$), and de-identification also removes the

TABLE V

OUT-OF-FOLD FORECASTING PERFORMANCE ON 274 SCORE CLAIMS (164 PAPERS). AUROC: MEAN±SD OVER 20 REPETITIONS OF PAPER-GROUPED 5-FOLD CV, WITH PAPER-CLUSTERED BOOTSTRAP 95% CI OF THE SEED-AVERAGED PREDICTIONS. BRIER: MEAN OVER REPETITIONS (LOWER IS BETTER). BSS: BRIER SKILL SCORE VS. B0. CUTOFF "POST": USES INFORMATION AVAILABLE ONLY AFTER REPLICATION DATA COLLECTION (LEAKAGE ILLUSTRATION). SELECTED CONFIGURATIONS; GBT, TEXT-ONLY AND PERMUTED-GRAPH VARIANTS ARE IN THE ARTIFACT.

ID	Features	Cutoff	AUROC	95% CI	Brier	BSS
B0	Training-fold prevalence	strict	0.492±0.001	[0.37, 0.56]	0.2474	+0.0%
B1	Original-study metadata, LR	strict	0.627±0.017	[0.55, 0.71]	0.2409	+2.6%
B4	B1 + flat citation counts	strict	0.613±0.019	[0.54, 0.69]	0.2432	+1.7%
M1	B4 + graph structure	strict	0.624±0.018	[0.55, 0.70]	0.2406	+2.8%
M2	M1 + dependency correction	strict	0.620±0.020	[0.55, 0.70]	0.2406	+2.8%
G	Graph + dependency only	strict	0.591±0.027	[0.53, 0.69]	0.2485	-0.4%
$M2_\ell$	M2	launch	0.620±0.020	[0.55, 0.70]	0.2405	+2.8%
D	B1 + planned replication design	launch	0.625±0.019	[0.54, 0.71]	0.2429	+1.8%
D^*	D + achieved size/power	post	0.638±0.024	[0.58, 0.72]	0.2403	+2.9%
$M2_\infty$	M2	none	0.614±0.017	[0.53, 0.71]	0.2409	+2.7%

TABLE VI

LLM-AS-JUDGE VERIFIERS ON THE SAME CLAIMS. *Identified*: TITLE, CITATION, ABSTRACT AND ORIGINAL STATISTICS; *de-identified*: FIELD AND ORIGINAL STATISTICS ONLY. AUROC WITH PAPER-CLUSTERED 95% CI; Δ VS. B1 ON IDENTICAL CLAIMS AND RESAMPLES. PROBE: SHARE ANSWERING THAT THE PAPER WAS REPLICATED IN SCORE (*aware*) FOR REAL SCORE PAPERS VS. JOURNAL-MATCHED DECOYS THAT WERE NEVER REPLICATED, AND ACCURACY OF NAMED OUTCOMES (COUNTED WHENEVER AN OUTCOME IS NAMED, REGARDLESS OF STATED AWARENESS) VS. THE MAJORITY-OUTCOME BASELINE ON THE SAME CLAIMS. BRIER (PLATT): AFTER PLATT SCALING FITTED ON TRAINING FOLDS ONLY (PAPER-GROUPED CV, 20 REPETITIONS). B1 HERE IS SCORED ON ITS SEED-AVERAGED PREDICTIONS, SO ITS BRIER DIFFERS SLIGHTLY FROM THE PER-SEED MEAN IN TABLE V.

Model	Input	AUROC	95% CI	ΔAUROC vs. B1	95% CI	Brier	Brier (Platt)	Top-10% acc.
Claude Haiku 4.5 (2025-07)	identified	0.566	[0.48, 0.65]	-0.061	[-0.154, +0.026]	0.2541	0.2466	0.52
	de-identified	0.592	[0.51, 0.68]	-0.036	[-0.132, +0.057]	0.2561	0.2479	0.70
Claude Opus 5.5 (2026-06)	identified	0.655	[0.57, 0.74]	+0.027	[-0.044, +0.103]	0.2392	0.2323	0.89
	de-identified	0.620	[0.53, 0.70]	-0.007	[-0.097, +0.077]	0.2478	0.2393	0.70
GPT-5.4 mini (2025-08)	identified	0.600	[0.52, 0.68]	-0.028	[-0.106, +0.050]	0.2753	0.2438	0.70
	de-identified	0.570	[0.48, 0.66]	-0.058	[-0.160, +0.038]	0.2758	0.2473	0.56
GPT-6.1 sol (2026-04)	identified	0.663	[0.58, 0.74]	+0.036	[-0.031, +0.105]	0.2594	0.2313	0.78
	de-identified	0.612	[0.53, 0.70]	-0.016	[-0.107, +0.073]	0.2627	0.2413	0.74
B1 (metadata LR)	same claims	0.628	[0.55, 0.71]	–	–	0.2387	–	0.44

Memorization probe

Model	Aware (real)	Aware (decoy)	Named outcomes	Named acc.	Majority
Claude Haiku 4.5 (2025-07)	0.0%	0.0%	0/274	–	–
Claude Opus 5.5 (2026-06)	0.0%	0.0%	3/274	0.67	0.67
GPT-5.4 mini (2025-08)	42.7%	24.0%	77/274	0.64	0.56
GPT-6.1 sol (2026-04)	2.2%	2.0%	6/274	1.00	0.83

abstract, which is legitimate information. Both rarely recall outcomes explicitly: they name outcomes for only 6 and 3 of 274 claims (GPT-6.1 sol's 6 are all correct). Their gain from identification therefore reflects either implicit memorization or a legitimate reading of the abstract, which cannot be separated from outside the model; R1 cannot be certified for these judges [31]. The probe also exposes a model-specific failure. GPT-5.4 mini claims knowledge of a replication for 42.7% ([37, 49]%) of real papers and 24.0% ([17, 33]%) of decoys, and names outcomes for 12 decoys that were never replicated: fabricated replication records. GPT-6.1 sol does so occasionally (2 decoys); both Claude models named none. Unlike instance-level contamination tests [44], the decoys calibrate a model's claimed recall against papers where any claimed outcome must be false. GPT-5.4 mini's excess awareness for real over decoy papers, with named-outcome accuracy above the majority baseline, is also consistent with partial genuine knowledge.

Confidence. Opus 5.5 is the only verifier whose top-decile gain has a CI excluding zero: 0.89 (CI [0.67, 1.00]) vs. 0.62, gain CI [+0.07, +0.38]. This is nominal (unadjusted across at least 11 selective comparisons) and was obtained on identified input, where memorization would act (de-identified: 0.70); for GPT-6.1 sol the gain CI ([−0.00, +0.36]) touches zero. The strongest candidate for gating is thus a judge whose temporal validity cannot be certified.

G. Robustness

Under three alternative success criteria from the SCORE package, B1 stays between 0.60 and 0.64 and M2 is never consistently better; weighting papers equally, as the SCORE analysis does, gives 0.632 (B1) and 0.602 (M2). Within fields, B1 is near chance in psychology and health (0.53, n=94); field estimates are descriptive.

TABLE VII

AGENT-IN-THE-LOOP TRIAGE ON IDENTICAL CLAIMS (SCENARIO A: 274 VALID QUERIES; B: 259 POST-LAUNCH QUERIES, WHERE THE UNPROTECTED TOOL SILENTLY RETURNS A NO-CUTOFF PREDICTION AND THE CONTRACT-AWARE TOOL REFUSES). U: AGENT SEES ONLY `p_replicate`; C: FULL CONTRACT OUTPUT. WRONG AUTO = ACCEPTED NON-REPLICATING PLUS REJECTED REPLICATING CLAIMS, AS A SHARE OF ALL CLAIMS; Δ = C$-$U WITH PAPER-CLUSTERED 95% CI. REVISED WORDINGS: TABLE IX.

	Agent	Esc. U	Esc. C	Wrong U	Wrong C	Δ wrong (pp)
A	GPT-5.4 mini	66%	93%	14.6%	3.6%	-10.9 [-16.4, -6.3]
	GPT-6.1 sol	81%	100%	9.1%	0.0%	-9.1 [-13.5, -5.1]
	Claude Haiku 4.5	59%	1%	17.9%	54.0%	+36.1 [+27.3, +44.4]
	Claude Opus 5.5	94%	100%	2.9%	0.0%	-2.9 [-5.2, -1.1]
B	GPT-5.4 mini	54%	100%	20.1%	0.0%	-20.1 [-29.0, -12.5]
	GPT-6.1 sol	75%	100%	12.0%	0.0%	-12.0 [-19.8, -5.6]
	Claude Haiku 4.5	49%	85%	21.6%	6.9%	-14.7 [-22.3, -7.5]
	Claude Opus 5.5	91%	100%	3.9%	0.0%	-3.9 [-7.5, -0.8]

VII. AGENT IN THE LOOP

Contract tests show that the tool *reports* its limits; they do not show that an agent *uses* them [36]. We therefore ran a minimal single-turn triage workflow. An LLM agent receives a claim and the output of `verify_claim`, and must *accept* the claim (the pipeline builds on it), *reject* it, or *escalate* it to a human reviewer with limited capacity. The system prompt is identical across workflows; only the tool output differs. The *unprotected* agent sees `p_replicate` alone; the *contract-aware* agent sees the full response. In scenario A, a valid query at the strict cutoff, both agents receive the leave-paper-out prediction for each of the 274 claims. In scenario B, a query dated 2023-06-01 after the replication launched, the unprotected tool silently returns the no-cutoff prediction, while the contract-aware tool returns its refusal; this covers 259 claims. Scenario B is deliberately asymmetric: it measures what an agent does when a contract violation is *silent* versus *surfaced*, not harm from leakage itself (the leaky prediction is no more accurate; §VI-C). All four LLMs served as agents (4,264 decisions, plus 5,191 in the control, robustness and revised-wording runs below; $11.10 in total).

Table VII shows the results. No verifier probability was extreme (range 0.30–0.76), so unprotected agents already escalated many claims. Their automatic decisions, however, were near chance and consisted mostly of rejecting claims that did replicate.

For three of the four agents, the contract information turned these errors into escalations: wrong automatic decisions changed by -10.9, -9.1 and -2.9 percentage points in scenario A (GPT-5.4 mini, GPT-6.1 sol, Opus 5.5), and these agents escalated every post-launch refusal in scenario B.

Control arms (Table VIII, scenario A). Without any tool, agents mostly escalate on their own (GPT-5.4 mini 100%, GPT-6.1 sol 100%, Haiku 4.5 98%). It is the bare probability that induces automatic decisions: relative to no tool, wrong decisions rose by $+14.6$, $+9.1$ and $+17.5$ points (CIs [$+9.8$, $+19.8$], [$+5.2$, $+13.5$], [$+11.9$, $+23.4$]). A generic "uncertain" note undoes part of this harm. The contract does more than that note for three agents, changing wrong decisions relative to the placebo by -5.1, -2.9 and -1.8 points (CIs [-9.2, -1.7], [-5.2, -1.1], [-3.6, -0.4]; unadjusted). Its effect is therefore not explained by discouraging text alone. But it beats *no tool* only for Opus 5.5 (-4.0, CI [-8.2, -0.8]), whose no-tool arm accepted some claims; an always-escalate policy matches it. For a near-chance verifier, the contract's value is to remove the false authority of a bare number, not to make automation better. That is a reduction in automatic errors, not an improvement in decision utility: if one escalation costs c relative to one wrong automatic decision, the contract-aware agent beats the unprotected one only when c is below 0.42, 0.47 and 0.47 for these three agents (for Haiku 4.5, never).

TABLE VIII

CONTROL ARMS FOR SCENARIO A (274 CLAIMS): SHARE ESCALATED / SHARE OF WRONG AUTOMATIC DECISIONS. *No tool*: THE AGENT SEES ONLY THE CLAIM. *Placebo*: `p_replicate` PLUS A GENERIC "THIS ESTIMATE IS UNCERTAIN" NOTE. *Contract*: FULL `verify_claim` OUTPUT. *Always escalate* NEEDS NO MODEL.

Agent	No tool	Placebo	Unprotected	Contract
GPT-5.4 mini	100% / 0.0%	82% / 8.8%	66% / 14.6%	93% / 3.6%
GPT-6.1 sol	100% / 0.0%	95% / 2.9%	81% / 9.1%	100% / 0.0%
Claude Haiku 4.5	98% / 0.4%	63% / 15.7%	59% / 17.9%	1% / 54.0%
Claude Opus 5.5	86% / 4.0%	96% / 1.8%	94% / 2.9%	100% / 0.0%
Always escalate	100% / 0.0%			

Prompt robustness. With an independently worded system prompt (three agents, scenario A), the ordering *bare number* $>$ *placebo* $>$ *contract* in wrong automatic decisions holds for both GPT agents. For GPT-5.4 mini the rates are 17.5%, 9.1% and 1.5%, and Haiku 4.5's misuse persists: it rejects 91% of claims, with 50.4% wrong. Haiku's placebo does not help (33.2% vs. 31.0% bare). Magnitudes are prompt-sensitive, however: GPT-6.1 sol's bare-number error rate falls from 9% to 1.8%.

The fourth agent, Claude Haiku 4.5, *misused* the contract. Shown the full response in scenario A, it rejected 96% of claims and its wrong automatic decisions rose from 18% to 54%. Its stated reasons show two misreadings: (i) it treated the verifier's weak skill ("AUROC 0.62") and the advice "do not gate actions on this score" as evidence *against the claim* rather than against the tool; (ii) it read the field `selective_ok` as a warning about "selective reporting". It also rejected 15% of the refusals in scenario B, as if a refusal were negative evidence. An exploratory revision (v2), designed *after* this failure, renamed the field `confidence_gating_supported`, made the advice action-oriented ("…escalate if a decision is needed") and stated that a refusal carries no information about the claim; Haiku 4.5 then escalated 100% of claims in both scenarios. Listing 1 shows the original wording, used in all main experiments. A correct tool is therefore not sufficient: contract fields and advice are themselves an interface that must be tested against the agents that consume them.

Pre-specified validation of R6 (Table IX). Because v2 was designed after the failure and tested on one agent, we froze

TABLE IX

PRE-SPECIFIED R6 VALIDATION (SCENARIO A, 274 CLAIMS; WRONG = WRONG AUTOMATIC DECISIONS). *R6*: WORDINGS V2–V4 UNDER TWO SYSTEM PROMPTS (OPUS: ONE), WORST CELL SHOWN. *Decision-ready*: A SYNTHETIC VERIFIER (AUROC 0.88) WHOSE V3 OUTPUT LICENSES ACCEPT/REJECT FOR CONFIDENT SCORES; *agree*: DECISIONS CONSISTENT WITH THE LICENSED ACTIONS.

	Orig.	R6	Decision-ready tool		
Agent	wrong	esc./wrong	bare: auto/wrong	R6: auto/wrong	agree
GPT-5.4 mini	3.6%	100%/0.0%	89%/15.0%	54%/4.7%	98%
GPT-6.1 sol	0.0%	100%/0.0%	84%/13.9%	53%/4.7%	100%
Haiku 4.5	54.0%	100%/0.0%	87%/15.3%	54%/4.7%	98%
Opus 5.5	0.0%	100%/0.0%	83%/12.0%	53%/4.7%	100%

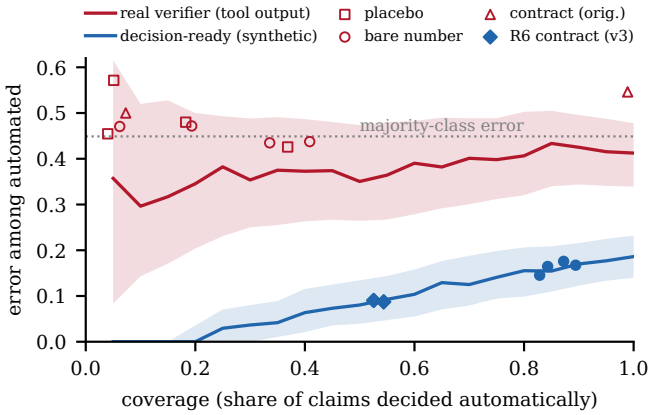


Fig. 1. Risk–coverage. Lines: threshold policy on each tool's scores (automate the k claims with largest $|p - 0.5|$, decide by $p \geq 0.5$; 95% bands). Points: observed agent decisions at self-chosen coverage, no budget imposed (hollow: real tool; filled: decision-ready).

two new wordings, the conditions, comprehension items and a support criterion, and committed them to the public repository before any call. v3 separates statements about the claim from statements about the tool and lists the licensed actions; v4 keeps the original fields, including `selective_ok`, and adds one interpretation sentence. All four agents saw v2–v4 under the main prompt, and the three agents of the robustness run also under prompt 2 (11,098 calls, $11.42). In all 21 agent–wording–prompt cells, agents escalated every claim, with 0.0% wrong automatic decisions, meeting the pre-specified criterion; Haiku 4.5's misuse disappeared under both new wordings (v3 vs. original: -54.0 points, CI $[-61.1, -46.3]$), and the three agents tested on refusals escalated all of them. Asked directly about the *original* output, every agent answered that the score cannot justify accepting or rejecting the claim, that the licensed action is escalation, and that a refusal carries no information about the claim (100% and 100% correct). Haiku 4.5 thus understood the output it misused; its only comprehension error was reading `selective_ok` as selective reporting (23% of answers; v4: 0%). The failure was a gap between understanding and acting, which stating the licensed action closes.

Not just abstention. Escalating everything is also what an always-escalate policy does. We therefore gave the same agents a synthetic *decision-ready* verifier, calibrated by construction (AUROC 0.88), whose scores outside $[0.15, 0.85]$ (53% of claims) are 91% accurate and whose v3 output licenses accepting or rejecting those claims and escalating the rest. With the contract, every agent automated all licensed claims and at most 4% of the others, agreeing with the licence on 98–100% of claims, with 4.7% wrong decisions. With the bare number, agents also automated 64–78% of unlicensed claims, with 12.0–15.3% wrong. Asked whether the score alone justifies a decision, agents distinguished licensed from unlicensed claims: 92% correct when scored per claim ('yes' for the 12 licensed, 'no' for the other claims of 30), applied uniformly to all agents. The frozen key wrongly expected 'yes' for all 30 (32% under it); no behavioural result uses either key, and both are in the artifact. The contract thus transmits *when* automation is warranted rather than merely discouraging it. This is a controlled interface experiment: it tests whether agents follow a licence, not whether a real verifier can earn one, and it says nothing about replication-forecasting accuracy.

Decision value under a review budget. With limited review capacity, some claims must be decided automatically, and the best use of a score is to automate the most confident ones. Fig. 1 evaluates that threshold policy on the scores; it is not an agent run under an enforced budget. For the real verifier, the error among automated decisions stays between 30% and 45% at every coverage: at escalation budgets of 25, 50 and 75%, 29.9%, 17.5% and 9.6% of all claims are decided wrongly, against 10.6%, 4.0% and 0.7% for the decision-ready tool. Agents given the bare real score sit at or above that curve. No wording can move the curve, since the contract adds no claim-level signal. Its value under a budget is to state, before deployment, that the budget rather than the tool must absorb the risk. The experiment remains minimal: single-turn decisions, one sample per decision, and a synthetic decision-ready tool.

VIII. DISCUSSION: IMPLICATIONS FOR AGENTIC SYSTEMS

Machine verifiers are weak signals, and apparently weaker than human crowds. Our best leakage-safe verifiers reach an AUROC of about 0.63 with a Brier skill below 3%. On SCORE, human forecasters reached a median AUC of 0.76 [21] (on different evaluation sets, so the comparison is indicative). An orchestrator should not prune hypotheses or literature on such a score; better uses are to prioritise claims for human or experimental checking and to surface the score with its base rate and evidence record. An LLM judge is the tempting replacement but not a drop-in one. The post-cutoff judges ranked best, yet part of their advantage may have depended on recognising the paper, and the cutoff they were told to respect cannot be enforced on their weights; the metamorphic test of §V has no analogue for knowledge held in parameters. Probability quality varied by model, and two models named outcomes for papers that were never replicated, so an agent that asks an LLM "did this replicate?"

can receive confident, false provenance. At minimum, LLM judges should be wrapped in the same contract: dated prompts, de-identified controls, decoy-calibrated memorization probes, base-rate reporting and audit logs, with recalibration before their probabilities are used.

Make temporal validity a type, and test it at the tool boundary. Most of the engineering effort went into enforcing R1–R3, not into modelling. Availability typing caught the achieved-size leak and the undated TOP Factor fields. Metamorphic tests at the tool boundary then provide regression-tested assurance that the deployed tool honours the cutoff the evaluation assumed. Agent tool descriptions should carry an explicit `as_of` parameter and an output schema with the base rate, evidence cutoff, gating verdict and provenance.

Add a consumer-side clause to the contract. The same output shifted three agents to escalation and misled a fourth, which read tool-quality warnings as evidence about the claim. We therefore propose a sixth requirement, **R6 Consumability**: the output must state the action it licenses (e.g., "escalate"), not only the actions it forbids, and the contract must be validated against the agents that will consume it. R6 is a *semantic* analogue of consumer-driven contract testing in service engineering [45]: there, consumers' expectations of a provider's interface are tested, while here LLM consumers' *interpretation* of the output is tested. Two pre-specified wordings satisfying R6 transferred to all four agents and, for three, to a second prompt, and let agents automate exactly what a decision-ready tool licensed (§VII).

Ship negative controls, contract tests and audits with the tool. The permutation null bounded the graph block's signal; mutation analysis exposed test gaps (and unreliable tool verdicts until re-run); a filesystem audit found two paper identifiers differing only in case (`RJb`, `rjb`) that had silently shared one cache entry (fixed throughout). Each check belongs in a verifier's continuous-integration suite, alongside data checksums and a label gate that reconciles to published totals.

IX. Threats to Validity and Limitations

Sample size. With 274 claims in 164 clusters, CIs are wide, and AUROC differences below roughly 0.04 cannot be resolved. Our nulls rule out large gains, not small ones. **Construct.** Replication success depends on the criterion, and a failed replication does not imply a false claim; we report three alternative criteria. **Scope.** SCORE samples social and behavioural science papers from 2009–2018 that were selected for replication. Results may not transfer to biomedicine, computational reproducibility or formal verification, which need different labels. Which claims SCORE chose to replicate is itself post-publication information that our cutoff does not control. **Evidence.** OpenAlex coverage and dates are imperfect, and author disambiguation errors affect dependency clusters. Our graph lacks semantic support/contradiction and dataset-reuse edges, so H1/H2 are tested only at bibliographic, paper-level granularity. **Models.** We evaluated deliberately simple, auditable models. The text model is a basic TF-IDF; richer encoders might extract more signal but bring contamination risk. We did not recalibrate the harness's own verifiers. The LLM judges were run with one prompt and the agents with two, one sample per claim, and each model's lowest reasoning setting, so reasoning settings differ across models. Two model identifiers are dateless, and knowledge cutoffs are vendor-stated. The v2 wording was designed after observing one agent's failure; v3 and v4 were pre-specified, but all wordings were written by the author. LLM results may change with prompts, sampling and versions; the probe measures only what a model will *say* it knows. Our intervals come from a paper-clustered bootstrap of seed-averaged out-of-fold predictions; refitting inside resamples widens B1's AUROC interval to [0.46, 0.72] (§VI-A). **Tool and agent.** The MCP tool is a research prototype. The agent experiment is a single-turn workflow with two system prompts, evaluated offline. Its controls cover scenario A only. R6 wordings say "escalate", so their 100% escalation partly reflects instruction following; the decision-ready control shows that agents also follow licences to act, but that tool is synthetic, with scores generated from the outcomes. The pre/post-cutoff split is confounded, because individual SCORE reports predate the consolidated paper. The methodology is demonstrated on one tool class; the schema and test relations are tool-agnostic, but applying them to a second tool is future work. H1–H3 were not preregistered; apart from the pre-specified R6 follow-up, analyses outside the Holm family are exploratory. Targeted mutants cover the five guarantees we implemented; the standard-tool analysis leaves 42 surviving mutants outside the contract's scope (§V). A mutation score measures how well the tests detect injected faults; it is not proof that the contract holds in every case.

X. Conclusion

The central lesson is methodological. A verifier tool for scientific agents should come with an epistemic contract, tests that enforce it at the tool boundary, and evidence that agents read it as intended. Each layer caught something the others missed: availability typing caught a leak in our own pipeline, mutation analysis exposed gaps in our tests (including at the cutoff boundary), and the agent experiment revealed a contract–consumer gap that no tool-level test can see. On the SCORE testbed, an integrity-compliant verifier discriminates only modestly, bibliographic evidence graphs yield no statistically established improvement, and confident predictions are not reliable enough to gate agent actions. The best LLM judges rank claims competitively, but their temporal validity cannot be certified, and two of the four judges named outcomes for papers that were never replicated. In the loop, a bare probability induced errors that most agents did not make without the tool; a contract that states the licensed action removed them for all four agents and, for a decision-ready tool, licensed automation that agents followed. Replication forecasts should inform scientific agents, not gate them.

Artifact Availability

A reproducibility package (code, MCP server, tests, provenance manifest, per-claim predictions, cached LLM and agent

responses, and the scripts that generate every table and in-text number) is publicly available at https://github.com/luxinlabs/scireligraph (MIT licence). SCORE data are available from OSF [7].

Acknowledgment

Portions of the data pipeline, analysis code and manuscript text were drafted with the assistance of an AI system (Claude, Anthropic). The author reviewed and verified all content; all numbers are generated from the released analysis outputs.

References

[1] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha, "The AI scientist: Towards fully automated open-ended scientific discovery," arXiv:2408.06292, 2024.

[2] L. Zheng *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, vol. 36, 2023, pp. 46 595–46 623.

[3] Open Science Collaboration, "Estimating the reproducibility of psychological science," *Science*, vol. 349, no. 6251, p. aac4716, 2015.

[4] C. F. Camerer *et al.*, "Evaluating replicability of laboratory experiments in economics," *Science*, vol. 351, no. 6280, pp. 1433–1436, 2016.

[5] ——, "Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015," *Nature Human Behaviour*, vol. 2, pp. 637–644, 2018.

[6] A. H. Tyner *et al.*, "Investigating the replicability of the social and behavioural sciences," *Nature*, vol. 652, no. 8108, pp. 143–150, 2026.

[7] Center for Open Science, "Materials for "Investigating the replicability of the social and behavioral sciences"," OSF project g5sny, analysis data and code package (15 Dec 2025), https://osf.io/g5sny/, 2025.

[8] U. M. Sehwag *et al.*, "SciPredict: Can LLMs predict the outcomes of scientific experiments in natural sciences?" in *Proc. ICML*, 2026, arXiv:2604.10718.

[9] A. Brodeur, D. Valenta, A. Marcoci *et al.*, "AI-assisted teams outperform AI-led teams but not human-only teams in assessing research reproducibility in quantitative social science," *Proceedings of the National Academy of Sciences*, vol. 123, no. 22, p. e2524747123, 2026.

[10] B. Meyer, "Applying "design by contract"," *Computer*, vol. 25, no. 10, pp. 40–51, 1992.

[11] M. Mitchell, S. Wu, A. Zaldivar *et al.*, "Model cards for model reporting," in *Proc. Conference on Fairness, Accountability, and Transparency (FAT*)*, 2019, pp. 220–229.

[12] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, 2017, pp. 1321–1330.

[13] T. Y. Chen, F.-C. Kuo, H. Liu *et al.*, "Metamorphic testing: A review of challenges and opportunities," *ACM Computing Surveys*, vol. 51, no. 1, pp. 4:1–4:27, 2018.

[14] Y. Jia and M. Harman, "An analysis and survey of the development of mutation testing," *IEEE Transactions on Software Engineering*, vol. 37, no. 5, pp. 649–678, 2011.

[15] Z. Wu, X. Liu, X. Zhang *et al.*, "MCPMark: A benchmark for stress-testing realistic and comprehensive MCP use," in *Proc. ICLR*, 2026.

[16] O. Sainz, J. A. Campos, I. García-Ferrero, J. Etxaniz, O. Lopez de Lacalle, and E. Agirre, "NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark," in *Findings of EMNLP*, 2023, pp. 10 776–10 787.

[17] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.

[18] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.

[19] M. Gordon, D. Viganola, A. Dreber, M. Johannesson, and T. Pfeiffer, "Predicting replicability—analysis of survey and prediction market data from large-scale forecasting projects," *PLoS ONE*, vol. 16, no. 4, p. e0248780, 2021.

[20] A. Dreber *et al.*, "Using prediction markets to estimate the reproducibility of scientific research," *Proceedings of the National Academy of Sciences*, vol. 112, no. 50, pp. 15 343–15 347, 2015.

[21] F. Mody *et al.*, "Large-scale human predictions of the replicability of published social and behavioural science papers – a multi-study analysis," MetaArXiv preprint, https://osf.io/preprints/metaarxiv/vgyed_v1, 2026.

[22] A. Altmejd *et al.*, "Predicting the replicability of social science lab experiments," *PLoS ONE*, vol. 14, no. 12, p. e0225826, 2019.

[23] Y. Yang, W. Youyou, and B. Uzzi, "Estimating the deep replicability of scientific findings using human and artificial intelligence," *Proceedings of the National Academy of Sciences*, vol. 117, no. 20, pp. 10 762–10 768, 2020.

[24] W. Youyou, Y. Yang, and B. Uzzi, "A discipline-wide investigation of the replicability of psychology papers over the past two decades," *Proceedings of the National Academy of Sciences*, vol. 120, no. 6, p. e2208863120, 2023.

[25] N. Alipourfard *et al.*, "Systematizing confidence in open research and evidence (SCORE)," SocArXiv preprint, https://doi.org/10.31235/osf.io/46mnb, 2021.

[26] S. Rajtmajer *et al.*, "Innovations in machine assessment of replicability," MetaArXiv preprint, https://osf.io/preprints/metaarxiv/bdmte_v1, 2026.

[27] T. Chakravorti, R. Fraleigh, T. Fritton, C. Griffin, V. Singh, S. Koneru *et al.*, "Human-AI collaboration for estimating scientific replicability," in *Proc. International Conference on Hybrid Human-Artificial Intelligence (HHAI)*, 2026, arXiv:2605.27394.

[28] B. Nguyen, D. Soós, Q. Ma *et al.*, "ReplicatorBench: Benchmarking LLM agents for replicability in social and behavioral sciences," in *Proc. ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2026, pp. 11 693–11 704.

[29] M. Serra-Garcia and U. Gneezy, "Nonreplicable publications are cited more than replicable ones," *Science Advances*, vol. 7, no. 21, p. eabd1705, 2021.

[30] D. Wadden, S. Lin, K. Lo, L. L. Wang, M. van Zuylen, A. Cohan, and H. Hajishirzi, "Fact or fiction: Verifying scientific claims," in *Proc. EMNLP*, 2020, pp. 7534–7550.

[31] D. Paleka, S. Goel, J. Geiping, and F. Tramèr, "Pitfalls in evaluating language model forecasters," in *Proc. ICLR*, 2026.

[32] Y. Zhang, N. Seedat, Y. Dong, P. Cui, J. Zhu, and M. van der Schaar, "GLEAN: Guideline-grounded evidence accumulation for high-stakes agent verification," in *Proc. ICML*, 2026, arXiv:2603.02798.

[33] T. Abbasli, K. Toyoda, Y. Wang, and L. Chen, "Claim-level confidence calibration for reliable decision making with large language models," in *Proc. 5th Workshop on Uncertainty Reasoning and Quantification in Decision Making (at ACM SIGKDD)*, 2026, arXiv:2608.22483.

[34] Z. Liu and Y. Zhang, "SGVEF-LOOP: Coverage-guided progressive topological exploration and fact-grounded metamorphic evaluation for MCP agents," in *Proc. ACL (Volume 1: Long Papers)*, 2026, pp. 26 573–26 599.

[35] S. Yergattikar, "Securing the tool layer: A threat taxonomy and runtime defense framework for model context protocol deployments," in *Proc. ACL (Volume 6: Industry Track)*, 2026, pp. 865–871.

[36] H. Hong, Y. Li, L. G. Huy, J. Yang, M. Zhou, and C. Lin, "Overview of the NLPCC 2026 shared task 11: Agent-based experiment reproduction from scientific papers," arXiv:2609.11117, 2026.

[37] D. Sculley *et al.*, "Hidden technical debt in machine learning systems," in *Advances in Neural Information Processing Systems*, 2015, pp. 2503–2511.

[38] S. Amershi *et al.*, "Software engineering for machine learning: A case study," in *Proc. ICSE-SEIP*, 2019, pp. 291–300.

[39] T. Gebru *et al.*, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.

[40] Anthropic, "Model context protocol specification," https://modelcontextprotocol.io, 2024.

[41] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.

[42] R. El-Yaniv and Y. Wiener, "On the foundations of noise-free selective classification," *Journal of Machine Learning Research*, vol. 11, pp. 1605–1641, 2010.

[43] J. Priem, H. Piwowar, and R. Orr, "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts," arXiv:2205.01833, 2022.

[44] S. Golchin and M. Surdeanu, "Time travel in LLMs: Tracing data contamination in large language models," in *Proc. ICLR*, 2024.

[45] I. Robinson, "Consumer-driven contracts: A service evolution pattern," https://martinfowler.com/articles/consumerDrivenContracts.html, 2006.